MyBrain: public research baseline

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MyBrain investigates how original evidence, accepted facts, obligations and authorized execution can remain connected in a system controlled by its user. Case management and a dependable daily overview are the first application candidates. This is a conceptual research synthesis, not an approved architecture, implementation specification or performance claim.

01 Publication scope and evidence limits

This public edition was synthesized from private research inputs. It is not a complete copy or a mechanically redacted version of those inputs. Original evidence, private source locations, names, administrative identifiers, exact dataset counts, personal infrastructure details and account-specific observations are intentionally excluded. No private source is linked from this document.

Statements below are research hypotheses, conceptual distinctions and proposed evaluation cases. No candidate system has been benchmarked here, no live source system has been independently audited for this edition, and no architecture or migration strategy has been approved. Vendor capabilities, versions, licensing and integration access require fresh primary-source verification before selection.

02 Objective

Investigate a user-controlled environment that connects knowledge with current commitments and actual execution. The environment should preserve its original evidence and allow its owner to export and restore authoritative data, change models and replace adapters without losing meaning.

Storage control, offline use, exportability and model independence are separate properties. The research must establish which are required for each capability and demonstrate them rather than infer them from a self-hosted deployment.

03 Conceptual distinctions

Concept	Research responsibility
Original evidence	Preserve source identity, content, version, attribution and relevant time information.
Statement or interpretation	Distinguish an explicit source statement, a user declaration and an AI inference.
Accepted domain state	Represent the currently accepted facts, statuses and applicable conditions with explicit rules.
Decision or correction	Preserve why a match, change or exception was accepted; do not silently undo it during reprocessing.
Derived knowledge	Rebuild summaries, search indexes and explanatory pages from authoritative records and evidence.
Authorized execution	Record the applicable mandate, intended action, attempt and observed external outcome.

Extraction confidence does not establish real-world truth. A proposal is not acceptance, a plan is not execution, and sending a message does not establish receipt, fulfillment or payment.

04 First application candidate

A case-management application could maintain related work items within a case and produce an evidence-based daily overview. A single case-level status may hide simultaneous obligations, dependencies and follow-up work. The research should test whether a small work-item model resolves that ambiguity without requiring a universal ontology.

A useful daily overview would identify relevant actions, dependencies due for reconsideration, new evidence and unresolved interpretation. Each entry should explain why it appears and link to an authoritative record and available evidence. A finished or superseded work item must not remain active merely because an older text summary still mentions it.

05 Time, identity and corrections

- Distinguish when something happened, when it was recorded and when a fact or agreement was valid.
- Keep deadlines, appointments and internal review dates separate; an expired review date is not automatically a missed external commitment.
- Preserve date precision rather than inventing a time for a date-only statement.
- Treat identity matches as revisable decisions supported by evidence; a matching string alone must not force an irreversible merge.
- Allow an accepted correction to survive re-import and model changes while retaining the original statement.
- Represent conflicting or newly discovered evidence explicitly rather than applying an unconditional latest-value-wins rule.

06 Ingestion and recoverability

Investigate a durable processing chain from received source version through interpretation to accepted changes. Advance a processing checkpoint only after the corresponding data has been stored durably. Repeated input must not duplicate evidence or effects, while a new interpretation must remain distinguishable from a new source event.

Source unavailability and synchronization gaps should be visible. A remote identifier is not by itself a recoverable source archive. Test restoring authoritative records and relevant source content without relying on the availability of the original connector. Search indexes and summaries should be reproducible from the retained authoritative material.

07 Authority and migration

Migration option	Research tradeoff
Existing system remains authoritative	Enables comparison with an established workflow, but the new copy has no independent write authority.
New system becomes authoritative, existing interface remains a projection	Requires explicit conflict handling and a controlled path for edits made through the old interface.
Existing system is replaced at transition	Requires sufficiently usable input, review, export and recovery from the first operational day.

An observation-only comparison is a candidate verification stage, not a permanent autonomy limit. No transition option is selected by this document. Determine the authoritative record per data type and agree the transition boundary before operational writes.

08 Mandates and uncertain outcomes

Autonomy should be defined through explicit, revocable mandates for an action type and data scope. A planned action must be checked against current conditions and authorization when it executes, including after a restart.

External outcomes need separate successful, failed and unknown states. If a timeout follows a possible external effect, reconcile the outcome before retrying. A durable internal workflow does not by itself establish that an external action happened exactly once. Instructions embedded in source content are data; they do not grant permissions or change the execution mandate.

09 Synthetic evaluation cases

These are fictional, deliberately generic scenarios for future evaluation. They do not identify or reproduce a private customer case.

Scenario	Proposed observable outcome
Completed work retains an old reminder in a summary	The accepted completion closes the work item; the obsolete reminder does not become new work.
Two independent obligations share one case	Each obligation has its own actor, conditions, status and review date.
An original message contains a subsequently corrected reference	Reprocessing preserves both the original statement and the accepted correction without a silent false merge.
A closed case receives a genuinely new request	New work is represented without erasing the prior completion or claiming an unrelated obligation is unfinished.
The same source batch is imported again	No duplicate source event or repeated external effect is created.
A process stops during ingestion	Recovery resumes from the last durable checkpoint and shows any unprocessed range.

Scenario	Proposed observable outcome
An action times out after possible completion	The result is marked unknown and reconciled before another attempt.
A mandate is revoked before a scheduled action	The action is re-evaluated and does not execute under expired authority.
A backup is restored without the original connector	Authoritative records and retained sources are readable; derived views can be rebuilt.

Expected outcomes need reference judgments before testing. Passing a small sample would not prove general correctness.

10 Open research decisions

- Define the smallest useful case and work-item semantics, including conditions, state transitions and time roles.
- Determine which record is authoritative for each kind of fact, and how conflicting edits are resolved.
- Specify source retention, correction, deletion and recovery requirements.
- Define mandates for internal changes separately from mandates for external actions.
- Choose storage and deployment according to write patterns, recovery requirements and operating effort; no database or hosting product is selected.
- Compare local, external and hybrid model processing on the same approved evaluation inputs and permitted data scope.
- Measure task accuracy, correction effort, missed or false work items, processing gaps, recovery and net maintenance effort.
- Select an initial operational scope before extending into additional personal or business domains.

11 Development governance

PGMCP is the development harness for research, design, planning, implementation and review. It is not the second brain's runtime memory, domain authority or execution authorization layer. Keep those responsibilities explicit.

Use real issues and registered issue branches for implementation. Record architectural choices separately from exploratory research. Public workflow state and deliverables must themselves be screened: making a state file versioned does not make private evidence suitable for publication.

Publication record

Prepared by Codex on 21 October 2026. Version 0.1 is a deliberately synthesized public research draft. Original private source PDFs are excluded from the repository workspace. The document records no approval of architecture, migration strategy or product readiness.